

SMART

SUSTAINABLE MANAGEMENT OF AGRICULTURE WITH THE INTELLIGENT COMPUTING CONTINUUM

Thematic Research Project submitted to FAPESP in response to the Call for Proposals FAPESP/NSFC – National Natural Science Foundation of China (https://fapesp.br/16944)

Principal Investigator: Prof. Carlos Alberto Kamienski

Santo André, September 2024.

Summary

Summary 2
General Information 3
1. Project Basis 4
1.1. Research Significance 4
1.2. Scientific Foundations 5
1.3. State of the Art 6
2. Research Objectives, Content, and Approach 8
2.1. Research Objectives and Key Issues 8
2.2. Research Content 9
2.2.1. Sensor Perception Integration 9
2.2.2. Terminal Computing 10
2.2.3. Intelligent Perception 11
2.2.4. Intelligent Decision-making 11
2.2.5. Sustainable Agriculture Management 12
2.3. Technical Approach 13
2.3.1. Data Collection via Air-Ground-Space Collaboration 13
2.3.2. Accurate and Stable Perception of Agriculture Production 14
2.3.3. Models for Sustainable Agriculture Management 15
2.3.4. Efficient and Autonomous Agriculture Terminal Devices 15
2.3.5. Sustainable Agriculture Management System 16
3. Workplan 16
4. Foundations for Cooperation 19
4.1. Complementarity of the Project Partners 20
4.1.1. Brazilian Scientific and Technological Strengths 20
4.1.2. Chinese Scientific and Technological Strengths 20
4.1.3. Technical complementarity between the two sides 21
4.1.4. Partner Contribution 22
4.2. Experimental Conditions 22
4.2.1. Brazilian Pilot 22
4.2.2. Chinese Pilot 23
References 24

General Information

Title

SMART: Sustainable Management of Agriculture with the Intelligent Computing Continuum

Coordinators

Brazil: Prof. Dr. Carlos Alberto Kamienski (Federal University of ABC – UFABC)
China: Prof. Dr. Lei Xue (Sun Yat-sen University)

Project Partners

Brazil

- Federal University of ABC (UFABC)
- Brazilian Agricultural Research Corporation (Embrapa)
- FEI University Center (FEI)
- University of Campinas (Unicamp)
- Pitaya Irrigation (Pitaya)

China

- Sun Yat-sen University (SYSU)
- Shenzhen Technology University (SZTU)
- Guizhou Aerospace Intelligent Agriculture Co. Ltd. (GAIA)

Abstract

Food security remains a significant global concern, particularly in large countries like Brazil and China. Modern agriculture faces the dual challenge of increasing crop productivity while ensuring environmental sustainability amid climate change. Precise and smart agriculture management is crucial, leveraging technologies like IoT, 5G/6G, and Artificial Intelligence to enhance decision-making, reduce resource usage, and minimize environmental impact. Innovative agriculture applications generate vast amounts of data from IoT sensors and remote sensing. However, the distributed nature of real-world infrastructures complicates data management. This is where the IoT Computing Continuum, which includes sensors, edge computing nodes, and cloud datacenters, comes into play. It handles the end-to-end data flow, a complex task that requires innovative solutions. Yet, ensuring a reliable dataflow for sophisticated models that enhance precision, food security, and environmental protection remains challenging. The SMART project aims to develop and deploy trustworthy platforms across the IoT computing continuum, from field sensors to the cloud. It will utilize various sensors, data sources, networking technologies, data management, and machine learning to ensure reliable dataflow and trustworthiness. The project will be piloted in Brazil and China, focusing on coffee and wheat crops. It will emphasize smart irrigation and fertilization to reduce water and fertilizer usage. This approach aims to contribute to sustainability goals and address the challenges of climate change.

1. Project Basis

1.1. Research Significance

Sustainable agricultural development is crucial to global food security, ecological and environmental protection, and economic growth. However, with the rapid increase in the global population, which is expected to approach 10 billion by 2050, agricultural production is facing unprecedented pressure. It not only needs to meet the growing demand for food but also needs to cope with the challenges brought by climate change, ensure food quality and safety, and maintain ecological balance [1]. There are many problems in the application of agricultural technology, which not only restricts the sustainability of production but also poses a severe threat to the environment and human health.

First, the lack of irrigation efficiency has led to a large amount of water waste, which threatens the sustainability of agricultural production and regional ecological balance. According to the FAO (Food and Agriculture Organization of the United Nations Global Information System on Water and Agriculture, global agricultural irrigation consumes about 70% of freshwater resources [2]. However, according to the International Water Management Institute (IWMI), the water efficiency of traditional flood irrigation systems is only 35%-40% [3]. This inefficient water resource management, not only due to over-irrigation but also due to backward irrigation technology and improper irrigation management, has caused water evaporation and underground leakage, further exacerbating the global water crisis and posing a considerable challenge to the sustainability of agricultural production and the balance of the ecosystem.

Secondly, the overuse of fertilizers and pesticides has caused severe environmental problems, further threatening the sustainability of agricultural production and ecological health. According to data from the Food and Agriculture Organization of the United Nations (FAO), global fertilizer use has increased from about 14 million tons in 1961 to nearly 20 billion tons in 2019. Excessive fertilizer application has led to eutrophication of water bodies, soil acidification, and heavy metal pollution [4]. Similarly, the abuse of pesticides not only causes pesticide residues and increased biological resistance but also harms non-target organisms, resulting in health problems and economic losses of billions of dollars each year due to environmental pollution. In addition, violations in agricultural production, such as the illegal use of banned chemicals and genetically modified seeds, undermine fair market competition and cause serious public health problems and environmental crises.

To address these challenges, the computing continuum, as an emerging technology system [5], has shown great potential in promoting sustainable agricultural development. The computing continuum technology brings real-time data collection, analysis, and application to the agricultural field by seamlessly integrating computing resources and services between physical and digital spaces, including edge computing, cloud computing, the Internet of Things (IoT), and Artificial Intelligence (AI) [6], significantly improving the intelligence of agricultural production and resource utilization efficiency.

1) The computing continuum can achieve real-time and geographical distribution of data processing through distributed computing, making the agricultural production process more intelligent and precise. For example, in the intelligent irrigation system, by real-time monitoring of soil moisture and crop water demand, the computing continuum technology can accurately control the

irrigation time and water volume, significantly improving the efficiency of water resource use [7]. Regarding fertilization and pesticide application, the computing continuum can accurately control the amount and application time of fertilizers and pesticides, reducing the negative impact on the environment and improving the utilization efficiency of fertilizers and pesticides, thereby ensuring the healthy growth of crops [8].

2) The computing continuum can process the massive data robots collect in real time through edge computing devices, including key indicators such as crop growth, soil moisture, and ambient temperature. These data are used to adjust the robot's working status and behavior strategy in real time through deep learning algorithms, thereby realizing precision agricultural management and improving operation efficiency and crop yield [9].

3) Based on the integration of these data, the computing continuum significantly improves the efficiency of agricultural resource utilization and shows great potential for promoting sustainable agricultural development. By integrating real-time monitoring data from soil, climate, and crops, the computational continuum can accurately control the amount and timing of water and fertilizer application, significantly reducing resource waste and the environmental burden of agricultural production [10].

Applying these technologies has revolutionized traditional agricultural production models and opened up a new path for sustainable agricultural development. In this technology-driven transformation, artificial intelligence and computational continuum technologies have shown significant advantages, not only improving agricultural production efficiency but also optimizing resource utilization and decision-making processes, providing a new strategic direction for sustainable agricultural management.

1.2. Scientific Foundations

To gain a deeper understanding of the role of the computing continuum in promoting the sustainable development of agriculture, the following evidence illustrates how the computing continuum can effectively address the complex challenges of modern agriculture by optimizing resource utilization, improving production efficiency, and enhancing decision support.

1) Real-time data collection and precise monitoring: Computing continuum technology enables real-time data collection of farmland environments through Internet of Things (IoT) devices. These devices can monitor key indicators such as soil moisture, temperature, nutrient levels, meteorological conditions, and crop growth status. This real-time data collection and precise monitoring provides agricultural managers with detailed farmland status information, allowing them to take precise agricultural intervention measures at the right time and location. This data-driven approach supports sustainable agricultural practices by significantly improving resource efficiency, reducing water, fertilizer, and pesticide waste, and reducing environmental burden.

2) Edge computing and distributed processing: Edge computing technology enables computing power to be distributed on agricultural equipment and field sensors, reducing data transmission delays and improving the response speed of agricultural management. By extending data processing from the centralized cloud to the network's edge, the vast amounts of data generated in farm fields can

be processed and analyzed locally. In this way, agricultural equipment can respond in real time and perform operations such as intelligent irrigation and precise fertilization, greatly improving the efficiency and accuracy of agricultural production. At the same time, distributed processing reduces dependence on the central server and enhances the robustness and security of the system.

3) Artificial Intelligence and Decision Optimization: Artificial intelligence technologies in the computing continuum can identify complex patterns and trends and perform predictive analysis by analyzing large amounts of agricultural data. Based on these analysis results, artificial intelligence systems can provide agricultural managers with optimized decision-making suggestions, such as crop rotation plans, pest and disease warnings, resource allocation optimization, etc. This data-driven decision support system can effectively deal with the uncertainty and complexity in agricultural production, reduce risks, and improve the sustainability of production.
4) Intelligent automation and autonomous control: Computing continuum technologies support agricultural robots, drones, and automated equipment that can autonomously complete field operations such as sowing, fertilizing, irrigating, and harvesting. By combining with real-time data and artificial intelligence systems, automated equipment can make autonomous decisions and operations, ensuring that every agricultural activity is carried out in the most efficient manner. This intelligent automation not only reduces labor burden but also improves the accuracy and consistency of agricultural operations, further promoting the sustainability of agricultural production.

1.3. State of the Art

Smart agriculture represents a new form of growing crops, increasing productivity, and decreasing the use of precious resources (water, land, energy) and chemical substances that harm the environment [11]. The use of technology to add intelligence to agriculture still has a vast potential to generate significant advances. Through continuous technological innovation and application promotion, smart agriculture will gradually change the global agricultural landscape and become an essential pillar of agricultural development in the 21st century. Smart agriculture relies on agricultural sensors to monitor the environment and production factors in real time, such as meteorological conditions (e.g., precipitation, humidity, temperature), water resources (e.g., water level, flow), field conditions (e.g., soil moisture, temperature) and the operation of the irrigation system (e.g., pressure, flow) [12]. These data are stably and reliably transmitted through wireless or wired communication networks and then intelligently analyzed and processed by the agricultural Internet of Things platform [5]. Precision and efficient irrigation is an important part of water and fertilizer management in the production process of smart agriculture [13]. It is an important link in developing high-water-efficiency agriculture and improving agricultural water use efficiency on a large scale. Through monitoring, transmission, diagnosis, decision-making, and dynamic management of crop water, comprehensive use of natural precipitation and precision irrigation strategies, adaptive irrigation control based on the water demand of crops throughout the growth period is achieved to ensure that crops grow under the most suitable water conditions, achieving comprehensive benefits such as water saving, fertilizer saving, labor saving, and environmental protection [14][15].

Many agricultural developed countries have pushed smart agriculture to the world's leading level by using various intelligent equipment. These countries have innovated and implemented appropriate modern agricultural development models based on their own geographical and climatic conditions, and this model is in the process of continuous optimization and improvement. Through precise production management, human and material resources are saved, and production capacity and product quality are significantly improved.

With the continuous development of artificial intelligence, many areas have integrated deep learning technology to build smart agricultural projects such as smart irrigation [16][17]. Chen et al. [18] and Reddy et al. [18] use machine learning algorithms to issue alerts based on crop data to remind irrigation and support websites and mobile applications to facilitate monitoring. Mohapatra et al. [20] proposed a neuro-fuzzy system for the Indian agricultural environment, which can predict soil moisture and its difference from the current moisture value and use it as the input of the fuzzy logic system. Keswani et al. [21] proposed a neural network model to predict soil moisture at a certain point in the future and use fuzzy logic to implement the control system. Lenka et al. [22] designed an elastic backpropagation optimization technique to train a neural network pattern classification algorithm to predict soil moisture content. Torres et al. [23] proposed a system that uses data fusion to obtain better values from sensor networks for the Brazilian agricultural environment. The system uses two deep learning models, one for crop soil moisture and the other for evapotranspiration, to determine appropriate irrigation times. Nakanishi et al. [24] modeled the system affected by short-term plant water stress, used deep learning technology for image processing, and predicted appropriate irrigation time accordingly. Kashyap et al. [25] proposed using the LSTM deep learning algorithm to predict the soil volumetric moisture content, irrigation period, and spatial distribution of water required to irrigate cultivated land the next day.

With the continuous updating and iteration of large-scale data sources and environmental analysis tools, global attention to precision agriculture continues to grow. Remote sensing technology allows growers to collect, visualize and assess crop and soil health at all stages of production conveniently and cost-effectively and is one of the leading technologies for precision agriculture. Khanal et al. [26] used multiple machine learning algorithms to develop a model for predicting soil properties and corn yield suitable for North America. Peña et al. [27] developed a robust and fully automatic object-based image analysis program to analyze weed maps in corn fields in the Spanish region on a series of drone images. Jensen et al. [28] explored the causes of biological variation and the possibility of remediation using infrared images of wheat in Australia. Galloza et al. [29] used land imagers to statistically model crop residues in Indiana. Kerkech et al. [30] used drones to collect images and combined them with convolutional neural networks to detect diseases in French vineyards. Sugiura et al. [31] also used drones to collect images of potato growing areas in Japan to achieve the downstream task of eliminating infected plants. Gnädinger et al. [32] used drones and image processing technology to evaluate the final field emergence of corn in Germany.

In recent years, IoT technology has been used at different agricultural industry production chain levels to achieve a broader range of monitoring and control tasks. Santos et al. [33] proposed an IoT technology based on LoRa to predict soil moisture and temperature combined with smart sensors for the Brazilian agricultural environment. Chen et al. [34] provided an IoT agricultural platform to realize the connection between sensors and actuators. Muangprathub et al. [35] developed a WSN-based crop irrigation system that can water according to humidity sensors. Nawandar et al. [36] developed a

low-cost intelligent irrigation monitoring system based on MQTT and HTTP. Phupattanasilp et al. [37] used AR technology to directly overlay IoT data onto real-world objects, enhancing object interaction. Keswani et al. [38] combined the Internet of Things WSN node to achieve the prediction of soil water demand based on a neural network. Dagar et al. [39] combined soil pH, temperature and humidity, water quantity sensors, etc., to achieve crop and field monitoring.

With the development of cloud computing, a computing continuum has been proposed and widely used in various application scenarios [40][41][42]. At the same time, some agricultural models have also begun to move to edge computing to improve training performance. Namani et al. [45] proposed a smart drone algorithm that uses drone data combined with cloud computing to establish sustainable smart agriculture. Lee et al. [46] allocate some deep learning layers in the agricultural model to fog nodes to achieve computing power offloading. Tsipis et al. [47] pushed cloud elastic resources to the underlying fog network to dynamically balance the generated load to address potential environmental hazards of olive oil production. Baghrous et al. [48] based on fog nodes and LoRa technology, reduced the total delay incurred when transmitting data to the cloud for processing in agricultural monitoring. Gia et al. [49] implemented a system consisting of sensor nodes, edge gateways, LoRa repeaters, fog gateways, cloud servers, and end-user terminal applications to achieve large-area coverage for agricultural and breeding applications.

2. Research Objectives, Content, and Approach

2.1. Research Objectives and Key Issues

The SMART project aims to build an efficient, accurate, and sustainable agricultural supervision and management technology system based on the computing continuum in response to the urgent need for the integrated development of global smart agricultural supervision, combining the technological advantages of Brazil and China. A globally applicable agricultural data collection and processing solution is formed by developing a computing continuum perception integration technology with high compatibility, high reliability, and autonomous decision-making capabilities, integrating multisource information collection equipment from the air, sky, and ground with edge computing technology. At the same time, high-precision crop status intelligent perception technology and intelligent decision-making algorithm models are developed to achieve all-around monitoring and prediction of crop growth status and support precision agricultural practices such as water-saving irrigation, environmentally friendly fertilization, and green spraying. Specific goals include:

1) Build a computing continuum perception integration technology with high compatibility and reliability, achieve extensive support for protocol standards of different countries and regions, and ensure seamless integration of self-organizing networks and edge computing of high-altitude, low-altitude, and surface information collection equipment.
2) Develop terminal (end-device) computing technology with autonomous decision-making capabilities, reduce the burden of data transmission by realizing intelligent computing and decision-making in drones and other aggregation nodes, and improve the equipment's autonomous cruising and automatic fault identification and repair capabilities.

3) Develop high-precision intelligent crop status sensing technology, integrate multisource sensing information, realize accurate identification and prediction of crop growth, environmental, and pest conditions, and provide reliable data support for smart agricultural management.
4) Design intelligent decision-making algorithms and models to support sustainable green agricultural development, ensure the efficiency and environmental protection of agricultural activities such as irrigation, fertilization, and spraying, promote resource conservation and environmental protection, and provide technical support for the development of global green agriculture.

The implementation of the SMART project will focus on building a smart agricultural supervision and management system with global applicability, promote the digital and intelligent transformation of agricultural production, improve the efficiency and sustainability of agricultural production, and contribute to global food security and environmental protection. To achieve this goal, some key issues will be addressed:

1) Accurate fusion and perception of multisource heterogeneous data
2) Reliable and autonomous control issues under complex network structure and application environment
3) Accurate perception of farmland production status under the restriction of complex factors
4) Intelligent regulation and control of farmland crop growth under sustainable constraints

2.2. Research Content

2.2.1. Sensor Perception Integration

Perception integration technology with high compatibility and reliability, involving: a) Surface sensor self-organizing network technology; b) Air-space-ground coordination technology (satellites, drones, and ground); c) Computing continuum access technology. The SMART project is committed to developing a highly compatible and reliable computing continuum perception integrated technology for application and deployment in the agricultural field. This technology integrates a variety of equipment, such as high-altitude satellites, low-altitude drones, and surface information collection sensors. It combines edge computing technology to build an efficient multilevel data collection and processing system. However, the changeable climate and terrain in the deployment environment cause efficiency problems in data collection and transmission. To improve the collaborative working ability between edge devices and ensure the reliability of the computing continuum perception system, this project will study the following key technologies and methods:

a) Surface sensor self-organizing network technology: The SMART project intends to use a combination of sensor self-organizing network technology and graph neural network algorithms to achieve self-optimization and intelligent management of agricultural sensor networks. Dynamically adjusting network parameters and paths can effectively reduce the redundancy and interference of collected information during transmission and improve the accuracy and efficiency of data transmission.

b) Air-ground-ground collaborative technology: The SMART project plans to adopt air-ground-ground integrated technology and achieve efficient deployment at the cloud-edge level by combining multiple devices such as satellites, drones, and ground sensors. The project will use data-driven intelligent optimization algorithms to improve the performance of edge computing while ensuring real-time data processing security and effectively reducing energy consumption.

c) Computing continuum access technology: The SMART project intends to adopt the computing continuum technology to achieve efficient deployment at the cloud, fog, and edge levels and effectively manage computing resources and energy consumption. The project will design a set of intelligent deployment frameworks to improve the computing performance of each layer while ensuring real-time data processing security and optimizing algorithm configuration and resource allocation for specific needs in agricultural production environments.

2.2.2. Terminal Computing

For the SMART project, "terminal computing" means the set of IoT devices (things) that operate near the end-user and collectively perform data collection (or perception) activities. Terminal computing technology with autonomous decision-making ability, involving: a) Edge image processing technology; b) Autonomous cruise technology of drones; c) Automatic equipment fault identification and repair technology. Since the traditional centralized computing model faces problems such as limited computing resources, high network latency, and insufficient bandwidth when processing massive data, to solve the challenges of data transmission bottlenecks, insufficient real-time performance, and untimely anomaly detection and processing, this project intends to study computing continuum terminal computing technology with autonomous decision-making capabilities. This technology mainly focuses on realizing intelligent computing and decision-making (i.e., edge computing technology) on terminal computer nodes (such as drones, aggregation nodes, etc.), aiming to reduce the amount of data transmission, improve system response speed, and realize automatic detection and repair of anomalies, thereby enhancing the overall intelligence and reliability of the system. This project intends to study:

a) Edge data processing: To reduce the amount of data transmission and improve information processing efficiency, the SMART project intends to develop preprocessing algorithms suitable for different types of image data (such as images taken by drones, remote sensing images, and images collected by cameras), optimize image quality, extract key features, and remove redundant information. By performing efficient edge computing processing at the data source, it can not only significantly reduce the amount of transmitted data and reduce the network burden but also provide a high-quality data foundation for subsequent in-depth analysis and decision-making.

b) Autonomous UAV patrol: The SMART project will develop the ability of UAVs to plan paths autonomously, navigate with obstacle avoidance, and intelligently perform agricultural tasks in farmland environments. By combining machine learning algorithms with real-time data feedback, UAVs can autonomously collect information in complex farmland environments and accurately perform operations such as pesticide application and fertilization based on the growth status of crops and environmental data.

c) Automatic equipment fault identification and repair: The SMART project will develop a fault monitoring algorithm based on edge computing to achieve real-time detection and classification of terminal node (such as UAVs, sensors, etc.) faults. At the same time, the research will explore intelligent solutions for fault repair, including automatic recovery of critical functions, switching to backup systems, or guiding manual intervention to ensure uninterrupted agricultural production and minimize production losses caused by equipment failures.

2.2.3. Intelligent Perception

Intelligent data perception (or collection) technology for highly accurate crop conditions. The main goal is to use AI models and to perform fusion at the edge in the collected multisource/multimodal data. By integrating the four perception technologies of crop growth environment, growth status, insect conditions, and yield expectations, and through the multimodal information fusion capabilities, more accurate and intelligent agricultural decision-making and control can be achieved, thereby improving agricultural production efficiency and crop quality.

a) Crop growth environment perception: Crop growth environment perception technology can monitor and analyze soil, climate, light, moisture, and other factors in real-time. These data help farmers accurately adjust agricultural activities such as fertilization, irrigation, and sowing, thereby increasing crop yield and quality and reducing yield losses caused by environmental discomfort. Reasonable use of water, fertilizers, and pesticides to avoid excessive or insufficient input. Identify and respond to extreme weather or adverse environmental conditions faster, adjust agricultural operations in time, and reduce the negative impact of climate change.

b) Crop growth status perception: It can monitor crop health, growth, nutritional needs, and other information in real-time. These data provide farmers with an accurate management basis. By monitoring crop growth status, abnormal crop growth or potential problems, such as water shortage, fertilizer shortage, or pests and diseases, can be discovered in time.

c) Crop insect situation perception: Crop insect situation perception technology can monitor and detect pest conditions in farmland in real time and detect early signs of pests. This technology can help farmers take measures early to prevent the spread of pests and reduce crop losses. Early warning systems can significantly reduce the threat of pests to crops and improve crop yield and quality.

d) Crop yield forecast perception: Through crop yield forecast perception technology, farmers and agricultural enterprises can more accurately predict the final yield of crops. Expected yield perception technology can also help farmers understand the growth status and yield potential of different crops so they can more reasonably allocate resources such as water, fertilizer, and labor.

2.2.4. Intelligent Decision-making

Involves the development of algorithms and models to support sustainable green agriculture, including:

a) Intelligent model for sustainable farmland production: This model will integrate and analyze a large amount of data from the agricultural environment to help make more accurate and efficient decisions. The model can consider various data, such as soil conditions, climate, and crop growth.

b) Intelligent decision-making and control of crop water-saving irrigation: This model will optimize crop water use, avoid waste, and achieve more efficient irrigation. This optimization is due to the model's ability to adjust irrigation methods according to the actual needs of crops and environmental conditions.

c) Intelligent decision-making and control of environmentally friendly crop fertilization: Excessive use of fertilizers increases production costs and can lead to other adverse effects such as groundwater pollution and soil degradation.

d) Intelligent decision-making and control of green spraying of crops: The model allows for more targeted and efficient application of pesticides and other chemicals, reducing unnecessary use and focusing only on areas or plants that need to be treated. Therefore, green spraying dramatically promotes sustainable agriculture development, minimizes agrochemicals' impact on the environment, protects biodiversity, maintains soil and air quality, and avoids contamination of food and water sources.

2.2.5. Sustainable Agriculture Management

It involves designing, implementing, and applying a sustainable agriculture management system based on a computing continuum. A sustainable agricultural management system based on a computing continuum can perfectly integrate air, sky, and ground perception data and provide intelligent support for agricultural production. The specific research contents are as follows:

a) Design of perception system architecture for air, space, and ground coordination. In modern agricultural management, accurate and comprehensive data collection is crucial. Therefore, this project proposes to build an air, space, and ground coordination perception system. The core goal of this system is to form a widely covered and data-rich agricultural monitoring network by integrating satellites, drones, and ground sensors. The design of this system reflects the current urgent need for the development of precision agriculture, aiming to optimize resource utilization and enhance crop production efficiency through real-time monitoring of all aspects of agricultural production. We can monitor an extensive range of farmland conditions through satellite technology, such as crop coverage and environmental change trends, which provide key data for macro-decision-making. At the same time, drones and ground sensors will compensate for satellites' shortcomings in temporal resolution and detail capture. Drones can conduct targeted high-resolution monitoring, while ground sensors provide real-time data on soil and microclimate. This integration of multisource data not only enhances the comprehensiveness and accuracy of the data but also provides a scientific basis for subsequent decision-making and management.

b) Design a sustainable management system for smart agriculture based on the computing continuum. In response to the current challenges of agricultural data processing, computing continuum technology provides an efficient solution. This project will develop a smart agricultural management system based on computational continuum technology, which will integrate and analyze massive

amounts of information from different data sources. The purpose of this system design is to achieve data-driven agricultural management. The system can predict crop growth trends through advanced analysis and machine learning technology, identify potential pests and diseases, and formulate effective prevention and control measures. For example, the system will use time series analysis to predict crop growth and combine image recognition technology to identify pests and diseases. In addition, the system will provide intelligent suggestions for irrigation and fertilization, optimizing these operations to reduce resource waste and increase crop yields. This smart agricultural management system improves the accuracy of agricultural production and is a key technical path to achieving sustainable development goals. The design and implementation of this system are the core of the entire research plan. It will directly affect the efficiency and effectiveness of agricultural management and support the actual operation of sustainable agriculture.

c) Safe and reliable adaptive management system deployment. In terms of system deployment, considering the complexity of agricultural management systems and the high requirements for stability, this project places particular emphasis on the security and reliability of the system. In the design, advanced encryption technology and security protocols will be used to protect the security of data transmission and prevent the leakage of sensitive information. At the same time, the system's hardware and software will adopt a modular and configurable design, enabling the system to adapt to various environmental conditions and facilitating future upgrades and maintenance. In addition, the system will also implement an automatic fault detection and recovery mechanism to ensure that the system can quickly resume operation in the event of hardware failure or data loss, minimizing potential production losses. This safe and reliable system design is the basis for long-term sustainable management. It ensures that the system can operate stably in various uncertain and challenging agricultural environments and supports the entire agricultural production process.

2.3. Technical Approach

The SMART project follows the idea of applying remote sensing and IoT technology in smart agriculture. It adopts the method of optimization and iteration of algorithm modeling and application modules to build a technical route of information collection, terminal-side intelligent management, farmland status perception, water, fertilizer, and pesticide intelligent control, prototype system construction, testing, and application. The achievement of these goals will employ the following technical approaches:

2.3.1. Data Collection via Air-Ground-Space Collaboration

The collaboration of data collection will involve conventional fixed sensors, flying sensors in drones, and remote sensing data collected by satellites.

- Autonomous and quick network organization involves two phases: designing a system networking framework and a dynamic topology solution. SMART will develop a comprehensive system architecture that connects end devices using LoRaWAN. It is particularly suitable for battery-powered devices and is widely used in IoT applications due to its low data communication rate, high receiving sensitivity, and excellent interference immunity. A key issue is how to effectively

build and maintain the dynamic network topology to ensure stability and efficiency when the nodes change dynamically, which is particularly important for LoRaWAN mesh networks [50]. SMART will approach this issue using graph neural networks (GNN) [51], which in recent years have become a cutting-edge technology for processing graph data due to its application in deep learning.

- Air-ground-space collaborative information collection involves deploying a cloud-edge framework and a bandwidth allocation algorithm. Traditionally, edge devices upload a large amount of data to the cloud, which leads to huge communication overhead and increases network delay. SMART will implement an integrated continuum collaborative framework by integrating multilevel data collection systems in the air and on the ground. Satellites perform large-scale monitoring and provide environmental data from a macro perspective, drones perform a more detailed local data collection, and ground equipment such as unmanned vehicles and sensor networks continuously monitor soil parameters and microclimate conditions. Filtering techniques based on an IoT trustworthiness framework will be used for that purpose [52]. Also, the bandwidth allocation between each upper-layer device and the lower-layer device is limited, which limits the data transmission rate. To address this problem, the project will develop a bandwidth allocation algorithm based on reinforcement learning.
- Management and monitoring system based on the computing continuum: the SMART project will adopt the IoT Computing Continuum (IoTinuum) with six stages [5] together with the IoT Architecture for Smart Applications (IoTecture) [40] so that we can precisely determine which architectural components (software, hardware, communication) will be deployed and their arrangement (Thing, Mist, Fog, Cloudlet, Cloud, App).

2.3.2. Accurate and Stable Perception of Agriculture Production

Due to the complexity and diversity of the terrain, landforms, and climate of the agricultural production environment in traditional agricultural production for a specific crop, farmers often rely on past planting experience to manage the crop. The SMART project will develop an accurate and stable farmland production perception platform.

- Crop growth environmental data based on multisource sensors: the project will collect environmental data such as air temperature, air humidity, wind speed, wind direction, solar radiation, UV intensity, CO2 concentration, oxygen concentration, pH value, soil temperature, soil moisture, and soil nutrients through different sensors including UV, humidity, temperature, light intensity, and soil moisture. Also, crop growth status data is based on image information: a combination of UAVs, ground cameras, and multispectral cameras to collect plant growth status information, such as leaf area index, crop height, and canopy coverage.
- Pest monitoring data based on air-ground multisource information: In the aerial detection of pest detection, this module uses a UAV thermal infrared camera and a high-resolution RGB camera to capture pest information over a wide area. The RGB camera captures visible light images of the crop surface to identify noticeable pest features, leaf spot areas, and pest symptoms. Thermal infrared cameras detect temperature anomalies on the crop surface, where elevated temperatures indicate plant damage due to insect infestation.

- Multimodal Farmland Sensing Algorithm: A multimodal farmland perception algorithm is split into three parts: crop growth environment identification and prediction, crop growth state identification and prediction, and crop pest condition identification and prediction, which analyzes the farmland data at different levels and time intervals.
- Full-cycle crop growth status identification and prediction: involves data collection during the entire growth cycle from sowing to harvesting. In addition to environmental data, crop growth data such as vegetation indices (NDVI, EVI), crop height, leaf area index (LAI), etc., need to be collected. These data are regularly acquired using UAVs and multispectral and hyperspectral cameras and are subjected to feature extraction.

2.3.3. Models for Sustainable Agriculture Management

Intelligent regulation of sustainable farmland water, fertilizer, and pesticide is integral to optimizing water-saving irrigation, environmentally friendly fertilizer, and green pesticide application. This process requires a combination of ground sensors, remote sensing data, and crop growth models to systematically manage the growing environment of crops through algorithms and rules to ensure rational use of resources and maximize crop yield and quality. SMART will build a green management model to realize the intelligent regulation of water, pesticide, and fertilizer integration.

- Decision rule system for water, pesticide, and fertilizer. Constructing a crop growth model to simulate the growth process and predict the growth results under different management strategies can help farmers and agricultural managers make scientific decisions according to various environmental conditions and management strategies to improve crop yield and resource utilization efficiency.
- Algorithms and Models for Sustainable Farmland Management: multiple data sources must be combined with diverse models based on the algorithms mentioned above. These models deal with irrigation, fertilization, and pesticide application.

2.3.4. Efficient and Autonomous Agriculture Terminal Devices

The SMART project will contribute comprehensively improve the level of intelligent agricultural management:

- Lightweight and efficient terminal data processing: end-side devices (i.e., terminal), such as drones, sensors, and cameras, play a key role in not only processing and analyzing large amounts of data in real-time but also improving the overall response speed of the system by optimizing data transmission and reducing bandwidth consumption. This activity involves adaptive filtering and noise reduction processing, data compression based on deep learning, and feature extraction and data screening.
- Autonomous UAV cruise algorithm: integrates multisource data, dynamic mission planning, autonomous control, and real-time adjustment functions to optimize operations in complex agricultural environments. This activity involves multisource data fusion and preprocessing, mission planning and path generation, and autonomous cruise control and real-time adjustment

- Efficient and accurate equipment fault identification, localization, and repair: Based on the above fault detection and localization methods, intelligent and autonomous fault resolution methods can ensure that the system can quickly return to regular operation when a fault occurs, minimizing the impact on production or service. This activity involves multi-dimensional data acquisition and fusion, as well as fault diagnosis, real-time data analysis, and accurate fault location.
- Intelligent and autonomous fault resolution algorithm: an intelligent and autonomous fault resolution method can ensure that the system can quickly resume normal operation when a fault occurs, minimizing the impact on production or service. The following are the key steps of the intelligent and autonomous fault resolution algorithm: autonomous repair decision, continuous learning and optimization, and remote monitoring and collaborative processing.

2.3.5. Sustainable Agriculture Management System

A sustainable and smart farmland management system based on the computing continuum incorporates the latest information technology, from real-time data collection to efficient data processing to intelligent decision-making execution.

- System design and technical architecture: The design concept of the system is based on the computing continuum - a multistage computing architecture that integrates cloud computing, fog computing, and edge computing, aiming at decentralization of data processing and real-time response to tasks. The system architecture consists of three main components: a data acquisition stage, a data processing stage, and an execution control stage.
- Technology implementation and innovative applications: Involve a group of activities related to the implementation of multisource data fusion, real-time data stream processing, intelligent decision-making algorithms, automated execution mechanisms, system safety and reliability, and data security and privacy protection (e.g., data deidentification), fault tolerance and data backup, and concerns about scalability and system flexibility (e.g., using microservices).

3. Workplan

The SMART research content and approaches presented in Section 2 comprise a workplan, which also spans a schedule for activities. The workplan comprises seven workpackages (WP) further divided into tasks, with a duration of 36 months. Five WPs represent scientific and technical activities, with a WP for communication, dissemination, and exploitation and a WP for project management activities. SMART will employ AI (artificial intelligence) techniques, mostly machine and deep learning, to address the identified researcher challenges in smart agriculture.

WP 1 Sensor Development, Integration, and Evaluation

Description: This WP comprises activities related to the development, use, integration, and evaluation of sensors, also including drones as flying sensors

Duration: M1 to M30

Partners: Embrapa, Pitaya, SZTU, GAIA

Task 1.1 Development and Evaluation of Fixed Soil Sensors

Task 1.2 Cosmic Ray Neutron Sensors

Task 1.3 Comparison of Sensor Technologies

Task 1.4 Drones as Sensing Devices

Task 1.5 Sensor Intelligence: AI in Agriculture Sensors

WP 2 Computing Continuum for Smart Agriculture

Description: This WP comprises all activities related to the computing continuum involving networking, computing, and data management aspects

Duration: M1 to M36

Partners: UFABC, SYSU, FEI, Unicamp

Task 2.1 IoT (LPWAN/5G/6G) wireless technologies and IoT trustworthiness

Task 2.2 Self-organized IoT Networks for Smart Agriculture

Task 2.3 IoT Management Platform: Monitoring and management of smart agriculture infrastructure, computing continuum, and applications

Task 2.4 Smart Agriculture application development and deployment for the Computing Continuum

Task 2.5 Continuum Intelligence: using AI for designing, developing, and optimizing the IoT Computing Continuum

WP 3 Remote Sensing for Smart Agriculture

Description: This WP involves all activities

Duration: M1 to M36

Partners: Embrapa, SYSU, SZTU, GAIA

Task 3.1 Satellites for Remote Sensing: selection and analysis

Task 3.2 Image processing for sustainable agriculture

Task 3.3 Crop Forecast and Modeling

Task 3.4 Sensing Intelligent: AI for remote sensing and crop forecast

WP 4 Smart Models for Smart Agriculture

Description: This WP comprises all activities related to the development of algorithms, machine learning models, and optimization techniques for smart agriculture

Duration: M7 to M24

Partners: UFABC, Embrapa, FEI, SYSU

Task 4.1 AI models and optimization algorithms for smart irrigation

Task 4.2 AI models and optimization algorithms for smart fertilization

Task 4.3 AI models and optimization algorithms for smart pest control

WP 5 Smart Agriculture Use Cases

Description: This WP generates demands and provides use cases for hands-on experiments

Duration: M7 to M36

Partners: UFABC, Embrapa, Pitaya, SYSU, GAIA

Task 5.1 Brazilian Pilot: Design, Development, and Evaluation of a Coffee Pilot in Brazil

Task 5.2 Chinese Pilot: Design, Development, and Evaluation of a Wheat/Vegetables Pilot in China

WP 6 Communication, Dissemination, and Exploitation

Description: Under this WP, project partners will take actions for the project results to be known by society (communication), by the scientific community (dissemination), and to be exploited by industrial partners (exploitation)

Duration: M4 to M36

Partners: UFABC, SYSU, Pitaya, GAIA

Task 6.1 Communication: activities related to making the project results available to a broader general non-technical audience

Task 6.2 Dissemination: scientific dissemination of project results

Task 6.3 Exploitation: usage of the developed technologies by the industrial partners (Pitaya and GAIA)

WP 7 Management

Description: This WP involves activities related to project administrative and technical management, as well as cooperation management (workshops and meetings)

Duration: M0 to M36

Partners: UFABC, SYSU

Task 7.1 Project Management: involves the administrative management of the project, considering activities of the researchers in both Countries

Task 7.2 Technical Management: involves the technical management of the project considering the collaboration

Task 7.3 Project Workshops: one in Brazil and one in China

Task 7.4 Virtual monthly meetings

Table 1 shows the SMART project schedule for 36 months divided in quarters for the seven workpackages and their tasks. The activities overlap since many partners will collaborate to the execution of the project.

Table 1: SMART Project Schedule

Activity	Years/Quarters											
	Year 1				Year 2				Year 3			
	Q1	Q	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10	Q11	Q12
WP1												
Task 1.1												
Task 1.2												
Task 1.3												
Task 1.4												
Task 1.5												
WP2												
Task 2.1												
Task 2.2												
Task 2.3												
Task 2.4												
Task 2.5												
WP3												
Task 3.1												
Task 3.2												
Task 3.3												
Task 3.4												
WP4												
Task 4.1												
Task 4.2												
Task 4.3												
WP5												
Task 5.1												
Task 5.2												
WP6												
Task 6.1												
Task 6.2												
Task 6.3												
WP7												
Task 7.1												
Task 7.2												
Task 7.3												
Task 7.4												

4. Foundations for Cooperation

The project team has a solid research foundation and rich practical application experience. Team members come from multiple disciplines, including computer science, agricultural science, environmental science, and engineering, with interdisciplinary research capabilities and a broad technical vision. In the past few years, the team has successfully developed and deployed several agricultural informatization projects and published relevant research results in international journals and conferences. In addition, the team

has close cooperative relations with multiple agricultural enterprises and research institutions, which provide valuable resources and support for this project.

4.1. Complementarity of the Project Partners

Project partners in Brazil and China form a consortium of universities, research institutions, and companies that complement each other and provide great potential to achieve high-impact results that can benefit both countries.

4.1.1. Brazilian Scientific and Technological Strengths

The Brazilian team is composed of three universities - the Federal University of ABC (UFABC), the FEI University Center, the University of Campinas – an agriculture research institution, the Brazilian Agricultural Research Corporation (Embrapa), and a company that provides irrigation solutions and is willing to use the project results, Pitaya Irrigation. Together, these partners have developed skills with the potential to benefit the SMART project:

1) Computing continuum: The Brazilian team has excellent capabilities in real-time analysis of large-scale agricultural data, especially in the computing continuum environment, which can fully utilize the processing power in the network. This enables the Brazilian team to process the large amount of IoT and remote sensing data the Chinese team provides instantly and respond and adjust quickly.
2) LoRaWAN technology: The Brazilian team has deep experience in the application of LoRaWAN technology, which can achieve data transmission in remote agricultural areas without relying on traditional network infrastructure. This technology will significantly enhance the coverage of the Chinese team's IoT system, allowing agricultural data in remote areas to be effectively monitored and managed.
3) Sensor development and evaluation: The Brazilian team has been working with sensor development for agriculture, such as soil moisture probes, and evaluating the use of cosmic ray neutron sensors for smart irrigation and remote sensing technologies for smart fertilization. The Chinese team can use this knowledge to streamline the use of different sensors for agriculture.
4) Smart irrigation and fertilization: The Brazilian team has extensive experience in smart irrigation and fertilization, using IoT sensors and algorithms to ensure precise irrigation. This improves the efficiency of water resource utilization and reduces waste and costs. Applying this technology will make the Chinese team's agricultural management more scientific and help both sides achieve efficient and sustainable agricultural production.
5) Machine and deep learning: The Brazilian team has experts in machine and deep learning who have been working for many years with smart agriculture models, for example, for water need estimation in irrigation, for identifying water spraying problems in center pivots, and for reducing data transmission over the IoT computing continuum. These specific skills related to artificial intelligence in agriculture can help the Chinese team accelerate technological developments.

4.1.2. Chinese Scientific and Technological Strengths

The Chinese team comprises two universities – Sun Yat-sen University and Shenzhen Technology University- and a company working with aerospace technologies for smart agriculture, the Guizhou Aerospace Intelligent Agriculture. Together, these partners have developed skills with the potential to benefit the SMART project:

1) Internet of Things technology: The Chinese team has extensive experience in Internet of Things technology and can deploy and manage complex sensor networks to achieve real-time monitoring of key agricultural parameters such as soil moisture, temperature, and vegetation growth. This will help optimize resource allocation, such as the efficient use of water resources, thereby improving the overall efficiency of agricultural production. The Chinese team's expertise on the Internet of Things will lay the technical foundation for the success of the cooperation project and provide data support for the Brazilian team.
2) Remote sensing technology: The Chinese team has deep technical accumulation in remote sensing monitoring of large areas, which can effectively identify differences in farmland and provide accurate data support. This capability can help the Brazilian team optimize agricultural inputs, such as the allocation of fertilizers and water resources, thereby maximizing crop yields.
3) Fertilizers and pesticides: In the agricultural field, the Chinese team has in-depth expertise in fertilizers and pesticides and can provide precise fertilizer and pesticide application recommendations to improve crop health and yield. This expertise will help the Brazilian team enhance the quality of crops and reduce pesticide waste.
4) Multimodal large model application: The Chinese team has advantages in applying multimodal large models, which can accurately predict climate patterns, identify potential crop disease warnings, and guide best planting practices. Through these models, the Brazilian team can improve the precision of agricultural production and avoid potential disaster impacts.
5) HarmonyOS: The Chinese team has a deep understanding of HarmonyOS and can apply this system to integrate and control agricultural equipment to create a highly automated and efficient agricultural ecosystem. This system will enable the Brazilian team to manage and coordinate agricultural resources more effectively, thereby improving productivity.

4.1.3. Technical complementarity between the two sides

For the SMART project, the technical advantages of Brazil and China are significantly complementary, which will provide a solid foundation for the successful implementation of the project. First, the Chinese team has rich experience in applying IoT and multimodal large models, which can realize real-time data collection and analysis of agricultural environments, especially accurate monitoring of soil moisture, climate conditions, crop growth conditions, etc. These technologies provide the necessary data support for intelligent irrigation and precise fertilization and pesticide application, ensuring the accuracy and efficiency of agricultural inputs. At the same time, the Chinese team's deep understanding of the HarmonyOS system can integrate various agricultural equipment to build an efficient and intelligent agricultural ecosystem, thereby optimizing the use of agricultural resources.

Secondly, the Brazilian team has performed well in computing continuum and intelligent irrigation technology, especially in real-time processing and analysis of large-

scale agricultural data. The Brazilian team is good at using computing continuum technology. By optimizing the allocation and sharing of computing resources, the data collected by the Chinese team can be instantly processed and provide accurate irrigation and fertilization decision support. In addition, the Brazilian team's expertise in LoRa technology and self-organizing networks can ensure the effective transmission and monitoring of data in remote agricultural areas and realize the intelligent management of large farmland areas. These technologies will further enhance the coverage and effectiveness of the Chinese team's Internet of Things system, making the entire smart agricultural supervision system more responsive and accurate.

Through the organic combination of the two sides' technologies, this project can achieve all-round supervision of smart agriculture, from data collection and analysis to practical application, comprehensively improve agricultural production efficiency and resource utilization, and provide strong technical support for the realization of sustainable agricultural development.

4.1.4. Partner Contribution

The SMART project is balanced regarding total effort, funds, instruments, equipment, skills, and access to pilot agriculture fields. The advantages of each side of the cooperation are described in Section 4.1, which will be distributed according to a pre-assignment described in Section 3 and following the research content and approach described in Section 2.

4.2. Experimental Conditions

The SMART project will have pilots in real farms in Brazil and China, which is already granted by previous cooperation among project partners and the farmers. All The project's technological deliverables (concepts, devices, algorithms, models, software/platforms) will be tested in at least one pilot.

4.2.1. Brazilian Pilot

The Brazilian primary pilot will be a plot of irrigated coffee by drip irrigation on the Nossa Senhora Aparecida da Água Branca farm (Figure 1), located in the rural area of Brotas, São Paulo state, which has a total area of 67 hectares of the Catuaí variety. The farm, located 68 kilometers from São Carlos (Embrapa Instrumentation's location), already participates in collaborative R&D activities with the SMART partner Pitaya. In addition, a secondary pilot for preliminary experiments will be Embrapa's LANAPRE (National Laboratory for Precision Agriculture) farm.



a) Farm overview



b) Close view of a producing coffee tree

Figure 1 – Nossa Senhora Aparecida da Água Branca farm in Brazil

4.2.2. Chinese Pilot

The Chinese primary pilot is a cotton farm located at the Naomaohu Town in Yiwu County in the Xinjiang region) and the pilot will cover an area of approximately 8,000 hectares. The farm needs an intelligent and precise application platform for irrigation, fertilization, and pest control to drive the modernization and sustainable development of agriculture, achieving cost reduction and efficiency increase, water saving, and production increase in cotton plantation in the Xinjiang region. The pilot will be operated by SMART partner GAIA (Guizhou Aerospace Smart Agriculture, which keeps an ongoing collaboration with the farm. A secondary pilot will be a vegetable farm located in the Guizhou Province.



a) Farm prepared for sowing



b) Cotton plantation

Figure 1 – Xinjiang Cotton farm in China

References

[1] Vågsholm, I., Arzoomand, N.S. and Boqvist, S., 2020. Food security, safety, and sustainability—getting the trade-offs right. Frontiers in sustainable food systems, 4, p.16.

[2] FAO. AQUASTAT: FAO's Global Information System on Water and Agriculture, 2016, 2016. Available online: http://www.fao.org/nr/water/aquastat/water_use (accessed on 24 August 2024).

[3] Provenzano, G. and Sinobas, L.R., 2014. Special issue on trends and challenges of sustainable irrigated agriculture. Journal of Irrigation and Drainage Engineering, 140(9), p.A2014001.

[4] Zahoor, I. and Mushtaq, A., 2023. Water pollution from agricultural activities: A critical global review. Int. J. Chem. Biochem. Sci, 23(1), pp.164-176.

[5] Kamienski, C., Zyrianoff, I., Bittencourt, LF and Di Felice, M., 2024, April. IoTinuum: The IoT Computing Continuum. In 2024 20th International Conference on Distributed Computing in Smart Systems and the Internet of Things (DCOSS-IoT) (pp. 732-739). IEEE.

[6] Togneri, R., Kamienski, C., Dantas, R., Prati, R., Toscano, A., Soininen, J.P. and Cinotti, T.S., 2019. Advancing IoT-based smart irrigation. IEEE Internet of Things Magazine, 2(4), pp.20-25.
[7] Kamienski, C., Soininen, J.P., Taumberger, M., Dantas, R., Toscano, A., Salmon Cinotti, T., Filev Maia, R. and Torre Neto, A., 2019. Smart water management platform: IoT-based precision irrigation for agriculture. Sensors, 19(2), p.276.
[8] Lavanya, G., Rani, C. and GaneshKumar, P., 2020. An automated low cost IoT based Fertilizer Intimation System for smart agriculture. Sustainable Computing: Informatics and Systems, 28, p.100300.
[9] Shaikh, T.A., Rasool, T. and Lone, F.R., 2022. Towards leveraging the role of machine learning and artificial intelligence in precision agriculture and smart farming. Computers and Electronics in Agriculture, 198, p.107119.
[10] Cyriac, R. and Thomas, J., 2024. Smart Farming with Cloud Supported Data Management Enabling Real-Time Monitoring and Prediction for Better Yield. In Intelligent Robots and Drones for Precision Agriculture (pp. 283-306). Cham: Springer Nature Switzerland.
[11] Tzounis, A., Katsoulas, N., Bartzanas, T. and Kittas, C., 2017. Internet of Things in agriculture, recent advances and future challenges. Biosystems engineering, 164, pp.31-48.
[12] Ullo, SL and Sinha, G.R., 2021. Advances in IoT and smart sensors for remote sensing and agriculture applications. Remote Sensing, 13(13), p.2585.
[13] Silva, A.O.D., Silva, B.A.D., Souza, C.F., Azevedo, B.M.D., Bassoi, L.H., Vasconcelos, D.V., Bonfim, G.V.D., Juarez, J.M., Santos, A.F.D. and Carneiro, F.M., 2021. Irrigation in the age of agriculture 4.0: management, monitoring and precision. Agronomic Science, 51, p.e20207695.
[14] Calbo, A.G., Vaz, C.M.P., Marouelli, W.A. and Porto, L.F., Tecnicer and Embrapa, 2019. Water tension sensor, system for characterizing and continuously measuring soil water, system for indicating critical soil water tension and irrigation rod. US Patent 10,365,196.
[15] Silva, CDOF, Grego, C.R., Manzione, R.L., Oliveira, SRDM, Rodrigues, G.C., Rodrigues, C.A.G., Speranza, E.A., Junior, A.L. and Koenigkan, L.V., 2024. Summarizing soil chemical variables into homogeneous management zones–case study in a specialty coffee crop. Smart Agricultural Technology, 7, p.100418.
[16] Togneri, R., dos Santos, D.F., Camponogara, G., Nagano, H., Custódio, G., Prati, R., Fernandes, S. and Kamienski, C., 2022. Soil moisture forecast for smart irrigation: The primetime for machine learning. Expert Systems with Applications, 207, p.117653.
[17] Togneri, R., Prati, R., Nagano, H. and Kamienski, C., 2023. Data-driven water need estimation for IoT-based smart irrigation: A survey. Expert Systems with Applications, 225, p.120194.
[18] Chen Y A, Hsieh W H, Ko Y S, et al. An ensemble learning model for agricultural irrigation prediction, 2021 International conference on information networking (ICOIN). IEEE, 2021: 311-316.
[19] Reddy K S P, Roopa Y M, LN K R, et al. IoT based smart agriculture using machine learning/2020 Second international conference on inventive research in computing applications (ICIRCA). IEEE, 2020: 130-134.
[20] Mohapatra A G, Lenka S K, Keswani B. Neural network and fuzzy logic based smart DSS model for irrigation notification and control in precision agriculture. Proceedings of the National Academy of Sciences, India Section A: Physical Sciences, 2019, 89: 67-76.
[21] Keswani B, Mohapatra A G, Mohanty A, et al. Adapting weather conditions based IoT enabled smart irrigation technique in precision agriculture mechanisms. Neural Computing and Applications, 2019, 31: 277-292.
[22] Lenka S K, Mohapatra A G. Hybrid Decision Model for Weather Dependent Farm Irrigation Using Resilient Backpropagation Based Neural Network Pattern Classification and Fuzzy Logic[C]//Proceedings of First International Conference on Information and Communication Technology for Intelligent Systems: Volume 1. Springer International Publishing, 2016: 291-302.
[23] Torres A B, da Rocha A R, da Silva T L C, et al. Multilevel data fusion for the internet of things in smart agriculture[J]. Computers and Electronics in Agriculture, 2020, 171: 105309.

[24] Nakanishi G, Mineno H. Examination of image features considering short-term changes in plant water stress[C]//2020 IEEE 9th Global Conference on Consumer Electronics (GCCE). IEEE, 2020: 743-744.
[25] Kashyap P K, Kumar S, Jaiswal A, et al. Towards precision agriculture: IoT-enabled intelligent irrigation systems using deep learning neural network[J]. IEEE Sensors Journal, 2021, 21(16): 17479-17491.
[26] Khanal S, Fulton J, Klopfenstein A, et al. Integration of high resolution remotely sensed data and machine learning techniques for spatial prediction of soil properties and corn yield. Computers and electronics in agriculture, 2018, 153: 213-225.
[27] Peña J M, Torres-Sánchez J, de Castro A I, et al. Weed mapping in early-season maize fields using object-based analysis of unmanned aerial vehicle (UAV) images. PloS one, 2013, 8(10): e77151.
[28] Jensen T, Apan A, Young F, et al. Detecting the attributes of a wheat crop using digital imagery acquired from a low-altitude platform. Computers and electronics in agriculture, 2007, 59(1-2): 66-77.
[29] Galloza M S, Crawford M, Heathman G C. Crop residue modeling and mapping using Landsat, ALI, Hyperion and airborne remote sensing data. IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing, 2013, 6(2): 446-456.
[30] Kerkech M, Hafiane A, Canals R. Deep leaning approach with colorimetric spaces and vegetation indices for vine diseases detection in UAV images. Computers and electronics in agriculture, 2018, 155: 237-243.
[31] Sugiura R, Tsuda S, Tsuji H, et al. Virus-infected plant detection in potato seed production field by UAV imagery//2018 ASABE Annual International Meeting. American Society of Agricultural and Biological Engineers, 2018: 1.
[32] Gnädinger F, Schmidhalter U. Digital counts of maize plants by unmanned aerial vehicles (UAVs)[J]. Remote sensing, 2017, 9(6): 544.
[33] dos Santos U J L, Pessin G, da Costa C A, et al. AgriPrediction: A proactive internet of things model to anticipate problems and improve production in agricultural crops. Computers and electronics in agriculture, 2019, 161: 202-213.
[34] Chen W L, Lin Y B, Lin Y W, et al. AgriTalk: IoT for precision soil farming of turmeric cultivation. IEEE Internet of Things Journal, 2019, 6(3): 5209-5223.
[35] Muangprathub J, Boonnam N, Kajornkasirat S, et al. IoT and agriculture data analysis for smart farm. Computers and electronics in agriculture, 2019, 156: 467-474.
[36] Nawandar N K, Satpute V R. IoT based low cost and intelligent module for smart irrigation system. Computers and electronics in agriculture, 2019, 162: 979-990.
[37] Phupattanasilp P, Tong S R. Augmented reality in the integrative internet of things (AR-IoT): Application for precision farming. Sustainability, 2019, 11(9): 2658.
[38] Keswani B, Mohapatra A G, Mohanty A, et al. Adapting weather conditions based IoT enabled smart irrigation technique in precision agriculture mechanisms. Neural Computing and Applications, 2019, 31: 277-292.
[39] Dagar R, Som S, Khatri S K. Smart farming–IoT in agriculture[C]//2018 International Conference on Inventive Research in Computing Applications (ICIRCA). IEEE, 2018: 1052-1056.
[40] Zyrianoff, I., Heideker, A., Silva, D., Kleinschmidt, J., Soininen, J.P., Salmon Cinotti, T. and Kamienski, C., 2019. Architecting and deploying IoT smart applications: A performance–oriented approach. Sensors, 20(1), p.84.
[41] Dustdar S, Pujol V C, Donta P K. On distributed computing continuum systems[J]. IEEE Transactions on Knowledge and Data Engineering, 2022, 35(4): 4092-4105.
[42] Pujol V C, Donta P K, Morichetta A, et al. Edge intelligence—research opportunities for distributed computing continuum systems. IEEE Internet Computing, 2023, 27(4): 53-74.
[43] Heideker, A., Ottolini, D., Zyrianoff, I., Neto, A.T., Cinotti, T.S. and Kamienski, C., 2020, November. IoT-based measurement for smart agriculture. In 2020 IEEE International Workshop on Metrology for Agriculture and Forestry (MetroAgriFor) (pp. 68-72). IEEE.

[44] da Costa Bezerra, S.F., Filho, A.S., Delicato, F.C. and da Rocha, A.R., 2021. Processing complex events in fog-based internet of things systems for smart agriculture. Sensors, 21(21), p.7226.
[45] Namani S, Gonen B. Smart agriculture based on IoT and cloud computing[C]//2020 3rd International Conference on Information and Computer Technologies (ICICT). IEEE, 2020: 553-556.
[46] Lee K, Silva B N, Han K. Deep learning entrusted to fog nodes (DLEFN) based smart agriculture[J]. Applied sciences, 2020, 10(4): 1544.
[47] Tsipis A, Papamichail A, Koufoudakis G, et al. Latency-adjustable cloud/fog computing architecture for time-sensitive environmental monitoring in olive groves[J]. AgriEngineering, 2020, 2(1): 175-205.
[48] Baghrous M, Ezzouhairi A, Benamar N. Smart farming system based on fog computing and lora technology[C]//Embedded Systems and Artificial Intelligence: Proceedings of ESAI 2019, Fez, Morocco. Springer Singapore, 2020: 217-225.
[49] Gia T N, Qingqing L, Queralta J P, et al. Edge AI in smart farming IoT: CNNs at the edge and fog computing with LoRa[C]//2019 IEEE AFRICON. IEEE, 2019: 1-6.
[50] Cotrim, J.R. and Kleinschmidt, J.H., 2020. LoRaWAN mesh networks: A review and classification of multihop communication. Sensors, 20(15), p.4273.
[51] Wu, Z., Pan, S., Chen, F., Long, G., Zhang, C. and Philip, S.Y., 2020. A comprehensive survey on graph neural networks. IEEE transactions on neural networks and learning systems, 32(1), pp.4-24.
[52] Junior, F.M.R. and Kamienski, C.A., 2021. A Survey on Trustworthiness for the Internet of Things. IEEE Access, 9, pp.42493-42514.